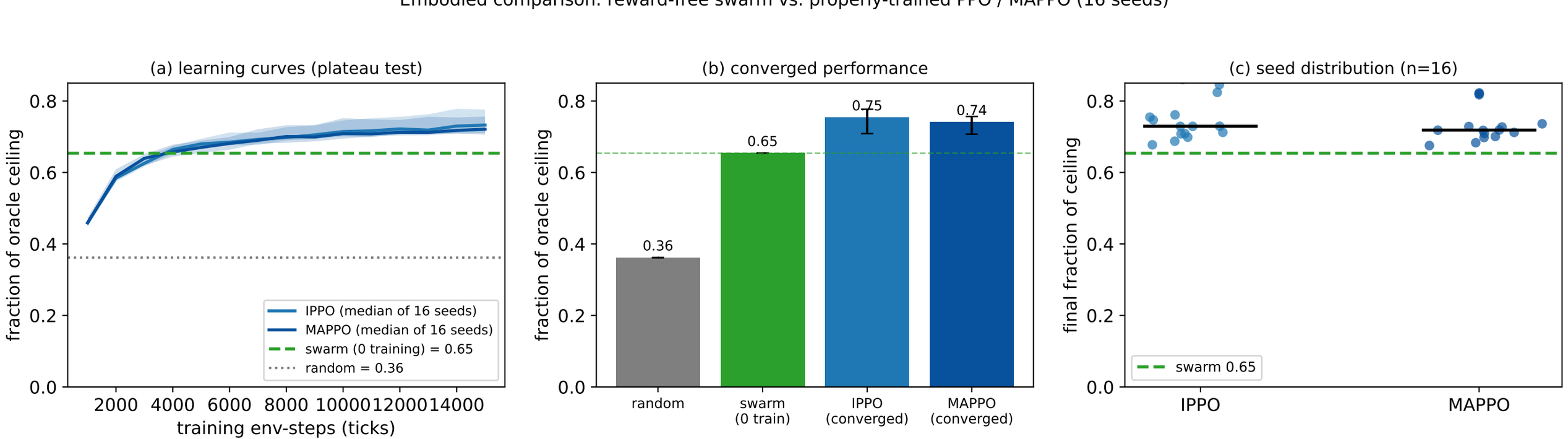
Embodied comparison: reward-free swarm vs. properly-trained PPO / MAPPO (16 seeds)
(a) learning curves (plateau test)
fraction of oracle ceiling
training env-steps (ticks)
IPPO (median of 16 seeds)
MAPPO (median of 16 seeds)
swarm (0 training) = 0.65
random = 0.36
(b) converged performance
0.36
0.65
0.75
0.74
random
swarm (0 train)
IPPO (converged)
MAPPO (converged)
(c) seed distribution (n=16)
final fraction of ceiling
swarm 0.65
IPPO
MAPPO